

HILAL YILMAZ

Production Planning & Manufacturing Digitalisation Engineer | Digital Twin

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Bursa, Türkiye | Open to relocation | Work permit / visa sponsorship required

PROFILE

Industrial & Management Engineer with 3 years in automotive serial production at Bosch and Tier-1 supplier Martur Fompak, working across production planning, capacity planning and manufacturing digitalisation. I build the data and tools behind planning decisions: Power BI KPI dashboards, and Excel-based capacity and workforce models that show whether lines, shifts and headcount can meet demand. Independent Researcher and invited speaker at DigiTwin 2026, University of Oxford, presenting a methods-engineering framework that translates factory KPI systems into digital twin models. MSc Management Engineering (Italy, full merit scholarship).

SELECTED RESULTS

- +7% availability, -8% cycle time and -10% scrap rate on Bosch CNC machining lines
- Monthly management reporting cut from ~2 weeks to 3 days for 4 production locations with one Power BI KPI system
- Invited speaker, DigiTwin 2026 (6th Digital Twin International Conference), University of Oxford

PROFESSIONAL EXPERIENCE

- Martur Fompak International** 05/2024 – 12/2025
Tier-1 automotive interior supplier | Bursa, Türkiye
- Method Improvement & Manufacturing Systems Development Specialist** 11/2024 – 12/2025
- Cut monthly management reporting from ~2 weeks to 3 days for 4 production locations, including Romania, by consolidating SAP, production, quality and logistics data into one Power BI KPI dashboard.
 - Gave management one consistent view of all lines, from plastic injection to door-panel and headliner assembly, by merging the separate Excel reports and differing KPI definitions of production, quality and logistics into a single data model.
 - Answered month-by-month capacity questions for plant managers (machine capacity limits, overtime need, headcount and labour cost) by building Excel scenario models for volume changes and project losses.
 - Made the dashboard the reference in strategic and holding-level management meetings by presenting findings and actions to plant, production, quality and logistics managers.
- Advanced Manufacturing Engineer** 05/2024 – 11/2024
- Delivered on-time SOPs with zero launch delays for Renault door-panel and trim programmes and Volkswagen, Skoda, Volvo and Tofaş projects by digitising and standardising PFMEA and process documentation with the process quality team.
 - Brought 6+ new lines to serial-production readiness, confirmed through run@rate and customer audits, by coordinating validation across quality, production, logistics and engineering.
- Bosch Powertrain Solutions** 02/2022 – 11/2022
Automotive powertrain components | Bursa, Türkiye
- Production & Planning Engineer**
- Raised availability by 7% and cut cycle time by 8% on CNC machining lines for valve and armature parts, tracked through line KPIs, by optimising machining processes.
 - Turned logistics demand into shop-floor production plans for 16 product variants (14 valve, 2 armature types) by combining SAP data with Excel-based capacity checks.
 - Reduced scrap rate by 10%, measured in monthly quality KPIs, by leading root-cause analysis and applying FMEA and SPC within Bosch's global quality framework.
- Co-founder, Business Analysis & Market Research – SoleLuna Tech** 11/2024 – Present
Software partnership building custom business management systems
- Shaped a web-based pre-accounting system for a retail and construction group by leading requirements and process analysis, designing the module structure, researching competitor solutions and running user tests.

PROFESSIONAL DEVELOPMENT & LEADERSHIP

- Leadership, Data & Digital Skills** 11/2023 – 07/2024
- Completed the 6-month Young Executive Academy (YEA; formerly YGA — Young Guru Academy) leadership programme and entrepreneurship training (11/2023 – 05/2024), while building Python skills for data analysis and starting German.
 - Completed the McKinsey Forward Programme, McKinsey & Company's structured development programme on problem solving and digital skills (01/2024 – 07/2024).

RESEARCH

Independent Researcher — Invited Speaker, DigiTwin 2026

08/2026

6th Digital Twin International Conference · University of Oxford, United Kingdom

- Presented a methods-engineering framework that translates KPI systems into digital twin models, built on KPI ownership in automotive serial production, and demonstrated that the framework generalises beyond manufacturing.

Research Project: Process Simulation & Optimisation

03/2018 – 06/2018

University of Salerno, Italy

- Modelled and optimised a production process with DELMIA and CATIA simulation, supported by statistical analysis in SPSS.

EDUCATION

MSc, Management Engineering — University of Salerno, Italy

2018 – 2021

Full merit scholarship

BSc, Industrial Engineering — Dumlupınar University, Türkiye

2012 – 2017

SKILLS

Production Planning: Capacity Planning · Workforce & Scenario Planning · Production Planning · OEE · Bottleneck Analysis · Shop-floor Management

Digitalisation & Data: Power BI · KPI Data Modelling · SAP (reporting & data extraction) · Advanced Excel · Python (data analysis, basic) · SPSS

Digital Twin & Industry 4.0: Digital Twin · KPI-to-Digital-Twin Framework · Industry 4.0 · Requirements & Process Analysis · Process Simulation (DELMIA 3/5 · CATIA 2/5, academic project)

Lean & Quality: Lean Manufacturing / TPS · 5S · Kaizen · Six Sigma · Value Stream Mapping · Standard Work · MTM-UAS · PFMEA / FMEA · Reverse FMEA · SPC · Root-Cause Analysis · Run@Rate & Customer Audits

CERTIFICATIONS & ONGOING TRAINING

- MTM-UAS & MTM Base — MTM Total Productivity (2025)
- Reverse FMEA — Rhythm Management Consulting (2025)
- AI Fluency — Anthropic Academy (2026)
- In progress: Microsoft Power BI Analyst Professional Certificate — Microsoft / Coursera
- In progress: Claude Code in Action (AI-assisted software development) — Anthropic Academy
- In progress: McKinsey.org Forward, Core Skills Level — McKinsey & Company (10/2026 – 12/2026)

LANGUAGES

Turkish: Native · English: Advanced (B2–C1) · Italian: Elementary (A2) · German: A2–B1 in progress, target B2 (Goethe certificate)